BareMinimum Healthcare Metrics and Implementation Guide

Metric definitions, evidence, data handling, and implementation steps

Purpose and scope

This guide explains the healthcare wellness and sustainability dashboard: what each number means, whether it is entered or calculated, what evidence supports comparisons, and how the application stores and protects records. The platform covers SDG 3 personal wellness, SDG 8 healthcare worker well-being, and SDG 12 healthcare resource and waste monitoring. Its personal records are dated and self-reported unless a future provider connection is explicitly added.

BareMinimum is a wellness and monitoring tool. It is not a medical device and does not diagnose or treat. Workload, safety, waste, and sustainability metrics are operational monitoring inputs; they do not create medical or regulatory determinations.

How to read the dashboard

- Recorded inputs are values typed into dated forms. They are labelled self-reported or manual and retain their entry timestamp.
- Calculated indicators use the recorded values and formulas described below. They are not independent measurements or clinical outcomes.
- Guidance comparisons come from WHO adult BMI and activity guidance, ICMR-NIN Dietary Guidelines for Indians (hydration context), ILO occupational safety principles, and WHO/CPCB healthcare waste guidance. They are comparison references, not individualized medical or compliance advice.
- The public landing page explains the workflow and metric catalog without collecting health data. The authenticated dashboard contains personal account records only. There are no sample personal measurements, aggregate scores, or generated demo values in the signed-in dashboard.
- Metric cards show units, recording period, last recorded date and source. Calculated indicators show formulas and their input record context. Blank values display as not recorded.
- If MongoDB is unavailable after sign-in, a device-only fallback is labelled local-only and namespaced to that account. It is not uploaded without a valid session.

Metric catalogue

The ranges below are input validation limits for this application, not recommended health or workplace limits. Empty fields are omitted from a saved record; a person must enter at least one value. Numeric values are validated in the browser and again on the server. Text fields are length-limited on the server.

SDG 3 Body measurements

Recorded metric	Meaning and source	Validation
Height	Measured value, entered by the user in centimetres.	0-250 cm
Weight	Measured value, entered by the user in kilograms.	0-500 kg
Waist circumference	Measured value, entered by the user in centimetres. No threshold is assigned by this app.	0-300 cm

SDG 3 Physical activity

Recorded metric	Meaning and source	Validation
Moderate activity	User-reported moderate-intensity minutes for the week.	0-1,000 min/week
Vigorous activity	User-reported vigorous-intensity minutes for the week.	0-1,000 min/week
Strength training	User-reported days with muscle-strengthening activity in the week.	0-7 days/week
Steps	Manual step count; no step recommendation is applied.	0-100,000 steps/day
Active time	Manual daily active minutes; the app does not infer intensity.	0-1,440 min/day
Sedentary time	Manual daily sedentary minutes.	0-1,440 min/day

SDG 3 Hydration

Recorded metric	Meaning and source	Validation
Water consumed	User-entered daily intake in millilitres.	0-20,000 ml/day
Estimated daily target	Optional user-entered estimate. The app supplies no universal fluid requirement.	0-20,000 ml/day

SDG 3 Sleep

Recorded metric	Meaning and source	Validation
Sleep duration	User-reported sleep duration. The current form compares it with general adult guidance but does not infer sleep quality or a disorder.	0-24 hours

SDG 3 Wearable-ready manual fields

Recorded metric	Meaning and source	Validation
Steps	Manually entered value; no wearable provider is connected.	0-100,000 steps
Heart rate	Manually entered beats per minute; not device-verified.	0-250 bpm
Active calories	Manually entered estimate; source and method are not verified.	0-10,000 kcal
Exercise duration	Manually entered duration.	0-1,440 minutes
Sleep duration	Manually entered duration.	0-24 hours

SDG 8 Healthcare worker well-being

Recorded metric	Meaning and source	Validation
Working hours	User-entered hours in a week.	0-168 hours/week
Shift duration	Reported length of a shift.	0-24 hours/shift
Consecutive shifts	Reported number of consecutive shifts.	0-31 shifts
Breaks taken	Reported number of breaks per shift and total duration.	0-30 breaks/shift; 0-1,440 minutes/shift
Overtime	Reported overtime hours in a week.	0-168 hours/week
Patients/tasks per shift	Reported workload counts.	0-1,000 patients/shift; 0-10,000 tasks/shift
Safety incidents	Reported safety incident count in the reporting period.	0-10,000 incidents/period
Needle-stick incidents	Reported needle-stick incident count in the reporting period.	0-10,000 incidents/period
Workplace injuries	Reported injury count in the reporting period.	0-10,000 incidents/period
Safety training completion	System-calculated from completed and assigned session counts.	0-10,000 sessions for each count
Wellness/fatigue assessment	Self-rating from 0 to 10. No diagnostic interpretation or risk category is assigned.	0-10

SDG 8 Timetable tasks

Recorded metric	Meaning and source	Validation
Task	Selected concrete measurement task, such as log water, record sleep, update activity, or record biomedical waste.	Required option from the defined task list
Category	Optional user-entered task category.	Up to 60 characters
Scheduled date	User-selected calendar date; no external calendar is connected.	Valid YYYY-MM-DD date
Planned duration	User-entered planned time for the task.	0-1,440 minutes
Task completed	User-selected planned/completed state.	Boolean

SDG 12 Biomedical waste

Recorded metric	Meaning and source	Validation
Total waste	Reported total biomedical/general waste for the day.	0-100,000 kg/day
Sharps	Reported sharps waste quantity.	0-100,000 kg/day
Infectious	Reported infectious waste quantity.	0-100,000 kg/day
Pharmaceutical	Reported pharmaceutical waste quantity.	0-100,000 kg/day
Chemical	Reported chemical waste quantity.	0-100,000 kg/day
General	Reported general waste quantity.	0-100,000 kg/day
Segregation checks	Reported total checks and correctly segregated checks.	0-100,000 records/day for each count
Recycled/recovered	Reported quantity recovered or recycled.	0-100,000 kg/day
Treatment/disposal records	Reported total treatment records and completed records.	0-100,000 records/day for each count
Waste handling incidents	Reported incident count.	0-10,000 incidents
Patients served	Patient count for the same reporting period, used as a denominator when available.	0-100,000 patients/day
Category/location notes	Optional user-entered context, up to 100 characters.	Text, 100 characters

SDG 12 Resource consumption

Recorded metric	Meaning and source	Validation
PPE	Reported personal protective equipment quantity.	0-1,000,000 items/day
Gloves	Reported quantity.	0-1,000,000 items/day
Masks	Reported quantity.	0-1,000,000 items/day
Syringes	Reported quantity.	0-1,000,000 items/day
Medicines	Reported medicine units; unit type is not normalized.	0-1,000,000 units/day
Packaging	Reported packaging mass.	0-100,000 kg/day
Electricity	Reported electricity use.	0-10,000,000 kWh/day
Water	Reported water use.	0-10,000,000 litres/day
Single-use items	Reported quantity.	0-1,000,000 items/day
Reusable items	Reported quantity.	0-1,000,000 items/day
Patients served	Patient count for the reporting period.	0-100,000 patients/day

SDG 12 Healthcare food waste

Recorded metric	Meaning and source	Validation
Food prepared	Reported prepared food mass.	0-100,000 kg/day
Food served	Reported served food mass.	0-100,000 kg/day
Food returned	Reported returned food mass.	0-100,000 kg/day
Food discarded	Reported discarded food mass.	0-100,000 kg/day
Meals served	Reported meal count.	0-1,000,000 meals/day
Cost of discarded food	Reported cost in Indian rupees.	0-100,000,000 ■/day
Patients served	Patient count for the reporting period.	0-100,000 patients/day

SDG 12 Sustainable procurement

Recorded metric	Meaning and source	Validation
Product and supplier	User-entered product and supplier labels.	Up to 120 characters each
Product cost	User-entered cost in Indian rupees.	0-100,000,000 ■
Packaging weight	Reported packaging mass.	0-100,000 kg
Recyclable packaging	User-reported percentage, not independently verified.	0-100%
Reusable/disposable status	User-entered product status.	Up to 30 characters
Expected waste	User-entered expected waste mass.	0-100,000 kg
Product lifespan	User-entered expected lifespan.	0-36,500 days
Transport distance	User-entered distance; route and method are not verified.	0-10,000,000 km
Supplier compliance	User-entered reported percentage; not a compliance certification.	0-100%
Treatment requirement	User-entered treatment text up to 160 characters and an optional mass field.	Text 160 characters; mass 0-100,000 kg

Calculated indicators and public-health comparisons

Indicator	Formula or comparison	Interpretation and limits
BMI	weightKg / (heightCm / 100)^2	Shown only when height and weight are both above zero. Adult screening bands are <18.5, 18.5-<25, 25-<30, and >=30. The form does not collect age; these adult bands are not suitable for children and BMI is not a diagnosis.
Weight change	newer weightKg - previous weightKg	Shown only when two dated body records contain a weight value. Both records remain visible in history; this is a difference between entries, not a target.
Moderate-equivalent activity	moderateMinutes + 2 x vigorousMinutes	WHO adult comparison guidance is 150-300 moderate minutes/week (or equivalent activity), plus strengthening on 2+ days/week. No capped progress percent is displayed.
Sedentary time	sedentaryMinutes / 60	Converts the user-entered daily sedentary minutes to hours/day; no recommended limit is assigned.
Hydration completion	waterMl / optional targetMl x 100	Shown only when the user enters a positive target. ICMR-NIN discusses approximately 2 litres including beverages for a general healthy person, with needs varying. This is context, not a personalized target.
Waste per patient	totalKg / patients	Shown only when the patient count is positive and comes from the same entry/reporting period.
Segregation accuracy	correctSegregationChecks / segregationChecks x 100	Shown only when the check count is positive. User-reported, not an inspection result. Correct count may not exceed total checks.

Indicator	Formula or comparison	Interpretation and limits
Treatment/disposal completion	completedTreatmentRecords / treatmentRecords x 100	Shown only when treatment record count is positive. This describes record completion, not regulatory compliance or treatment quality.
Tracked resources per patient	(PPE + gloves + masks + syringes) / patients	Combines only those four count fields. The units may not be comparable; it is a descriptive indicator, not a normalized efficiency score.
Food waste percentage	discardedKg / preparedKg x 100	Shown only when prepared mass is positive. Based on entered values.
Food waste per patient	discardedKg / patients	Shown only when patient count is positive.
Safety training completion	trainingCompleted / trainingAssigned x 100	Only shown when assigned count is positive. Completed may not exceed assigned; self-reported counts are not an audit.
Dated trends	Show saved dated measurements for the selected metric	Values are not aggregated into monthly totals unless the reporting period makes a sum meaningful. Record dates remain visible.
Worker monitoring inputs	Displays hours, shift length, break counts, workload and incidents	No fatigue threshold, medical risk score, or diagnosis is calculated.
Procurement fields	No composite score is calculated	Cost, packaging, lifespan, distance, and reported supplier values remain separate inputs. No SDG score or compliance percentage is invented.

Scientific basis and sources

Public guidance supports only the explicit comparisons listed above. The other metrics are operational data fields or arithmetic descriptions of the values a user enters. They do not use invented thresholds.

- WHO, Obesity and overweight (adult BMI): https://www.who.int/en/news-room/fact-sheets/detail/obesity-and-overweight
- WHO, Physical activity: https://www.who.int/initiatives/behealthy/physical-activity/
- ICMR-NIN, Dietary Guidelines for Indians 2024: https://nin.res.in/dietaryguidelines/pdfjs/locale/DGI_2024.pdf
- ILO, Occupational safety and health in workplaces: https://www.ilo.org/topics-and-sectors/occupational-safety-and-health-guide-labour-inspectors-and-other/workplaces
- WHO, Health-care waste: https://www.who.int/news-room/fact-sheets/detail/health-care-waste

- CPCB, Guidelines for management of healthcare waste as per BMWM Rules: https://cpcb.nic.in/uploads/projects/bio-medical-waste/guidelines_healthcare_june_2018.pdf

Healthcare waste labels follow the broad categories discussed by WHO (including sharps, infectious, chemical, pharmaceutical, and general waste). Local regulations and facility procedures determine actual handling, segregation, transport, and treatment; this application does not encode jurisdiction-specific compliance rules.

Technology stack and data flow

Layer	Technology and role
User interface	React 19 + TypeScript + Vite; the existing CSS theme and Lucide icons are retained. HealthcareWorkspace renders form fields from a small dataset catalogue.
Navigation and components	Existing Wouter routes, app page and Radix UI components remain. The new workspace uses the existing visual language and responsive CSS.
API	POST /api/auth/local/signup and POST /api/auth/local/login; GET /api/trpc/auth.me; POST /api/trpc/auth.logout (also POST /api/auth/local/logout). Protected: GET/PUT /api/dashboard and /api/health-record; GET/POST /api/healthcare-records; PUT/DELETE /api/healthcare-records/:id.
Authentication	Email/password accounts use Node scrypt with per-account salts. The browser receives an HttpOnly, SameSite=Lax cookie. MongoDB authSessions stores a SHA-256 token hash with expiry; authenticateRequest verifies the session and resolves openId on the server.
Database	MongoDB Node driver uses localAccounts, authSessions, dashboardSnapshots, healthRecords, and healthcareRecords. Dated records carry userId, dataset, values, recordedAt, and source. Reads, updates, and deletes include the verified account ID.
Local mode	Offline fallback is account-scoped in browser localStorage and labelled local-only. It is not uploaded without a valid session. The dashboard no longer displays sample personal measurements.
Validation	The dataset catalogue defines numeric ranges, text lengths, date formats, and boolean task state. The server repeats validation, rejects unknown fields, invalid dates, out-of-range values, and waste subtotals above total waste.

Implementation steps completed

- Reviewed the existing React dashboard, Express server, MongoDB helper, local account/session flow, and the BareMinimum real-world guide.
- Added a typed healthcare dataset catalogue with field labels, units, validation limits, text fields, and public guidance links.
- Added owner-scoped MongoDB listing, insertion, update, and deletion functions in the existing MongoDB module.
- Added MongoDB-backed account sessions with hashed password storage, revocable token hashes, expiry, authenticated profile status, and logout.
- Added authenticated REST endpoints. Each endpoint derives the user key from the verified session, not from request input.
- Added a public first-use workflow and exact metric catalog; no personal health information is needed to view it.
- Replaced the signed-in sample dashboard with Overview, SDG 3, SDG 8, SDG 12, Timetable and Trends / Reports pages.
- Added real metric cards with units, reporting frequency, last date, source, and formulas/guideline comparison where relevant; empty measurements say No data recorded.
- Restricted timetable tasks to concrete measurement prompts and validated the allowed task list server-side.
- Added account-scoped local fallback, loading/saving/error feedback, dated history, edit controls and formulas, without sample personal metrics.
- Added this metric and implementation handoff with source links and explicit limits for unsupported claims.

Running and verifying

Set MONGODB_URI and optionally MONGODB_DB_NAME in .env. On Windows PowerShell, run $env:PORT = "3001" followed by pnpm dev, then open http://127.0.0.1:3001. The cross-platform tools/run-server.mjs wrapper sets NODE_ENV, so cross-env is not required. For production, use pnpm build followed by pnpm start. pnpm check performs TypeScript validation; pnpm test runs Vitest; pnpm verify:auth runs the MongoDB-backed account isolation smoke test and removes its temporary records.

Verification performed: TypeScript validation passed; Vitest passed all 5 tests; Vite and Express production builds completed. The MongoDB end-to-end test passed sign-up, sign-in, health and dashboard reads/writes, timetable persistence, session persistence across server restart, logout returning protected access to HTTP 401, and two-account isolation from reads, updates, and deletes. Browser sign-up reached the authenticated dashboard and showed the account sync status. Temporary test accounts and records were removed. MongoDB ping succeeded. No wearable or external calendar provider is connected.

Current limits and next integration work

- No real wearable provider is connected; steps, heart rate, calories, exercise, and sleep values are manual entries.
- No healthcare facility, payroll, safety reporting, procurement, kitchen, scale, or waste-treatment system is connected. Operational data is self-reported/manual.
- No provider calendar or wearable integration is connected. Wearable-ready values are manual only.
- Facility operations and procurement inputs are user-entered and are not independently verified or regulatory determinations.
- Patient and worker roles are not currently separated by an authorization policy. Existing authenticated accounts own their own record sets; an organization-level management view requires a designed role/consent model before deployment.
- Email verification, password reset, and multi-factor authentication are not implemented. Add those controls and a rate limit before a public production launch.

Metric display and workflow

The unauthenticated page explains all three SDG areas and the exact available fields. Signed-in navigation includes Overview, SDG 3 Health & Wellness, SDG 8 Worker Well-being, SDG 12 Healthcare Sustainability, Timetable, and Trends / Reports. Overview lists configured inputs (including missing values) and available calculations. Metric cards show their unit, reporting period, last recorded date, source, and formula or comparison where applicable. The timetable offers concrete measurement tasks. Trends show dated saved entries for the selected dataset. Users can edit prior records; the new edit is timestamped and owner-scoped.